SEMESTER 5 – UNIT 1 – SCIENCE AND PURPOSE

1. Reading Comprehension

Science as a Mission, Not Just a Major

Every year, universities across Morocco welcome thousands of students into scientific disciplines such as biology, physics, chemistry, and geology. While many enroll simply because they were successful in scientific subjects during high school, only a minority fully understand the transformative power of science. Science is not just a collection of theories to memorize; it is a practical tool for solving problems.

Consider Agadir, a city facing high levels of plastic waste, water scarcity, and unemployment. A chemist might design biodegradable packaging to reduce pollution. A biologist could study drought-resistant plants in the Souss region to support local farmers. A physicist might invent affordable solar-powered systems for rural homes. These innovations do not emerge overnight; they result from persistence, teamwork, and a strong sense of responsibility.

Unfortunately, many science students limit themselves to attending lectures and preparing only for exams. They rarely join scientific clubs, research workshops, or national competitions. As a result, they earn degrees but lack real-world impact. To become true scientists, students must go beyond academic requirements. They should read scientific articles regularly, participate in experiments even when resources are limited, and collaborate with others to design practical solutions.

Science should not be pursued for the sake of prestige, but rather for the sake of progress. A nation does not advance through politicians alone; it advances through engineers who build systems, researchers who discover cures, and educators who spread scientific thinking. Morocco does not only need graduates; it needs innovators.

The question every science student should ask is not “What job will I get after graduation?” but rather “What problem will I contribute to solving?” When purpose drives learning, science becomes more than a subject; it becomes a mission.

A. Answer these questions

1. What is the main argument of the text? _____
2. Why does the author criticize some science students? _____

B. Are these statements True, False, or Not Mentioned?

Statement	T	F	NM
1. Science is only about memorizing theories.			
2. Agadir faces environmental and social challenges.			
3. Most science students in Morocco participate in research projects.			
4. The author believes scientists are more important than politicians.			

C. Multiple Choice

1. According to the text, real scientific contributions require:

A. High grades
B. Expensive laboratories
C. Persistence and teamwork
D. Foreign diplomas

2. What question should science students ask themselves?
    A. How much salary will I get?
    B. What problem will I help solve?
    C. Which subject is easiest?
    D. How many exams do I have?

D. Answer these questions:

1. Do you agree that many students “study only for exams”? Why or why not?
2. What scientific problem in Morocco interests you personally?

2. Vocabulary:

A. Match the Word to Its Meaning

Word	Meaning
1) scarcity	a) strong feeling of duty
2) innovation	b) lack of something needed
3) responsibility	c) new and creative idea
4) collaborate	d) work together

B. Word Formation

Complete the table:

Verb	Noun	Person
Innovate		innovator
Discover		
Educate		

C. Complete the Sentences

1. Agadir suffers from water _____ (scarce / scarcity / scarceful).
2. Scientists must _____ (collaborate / collaboration / collaborative) to find solutions.
3. Plastic pollution requires urgent _____ (innovative / innovation / innovating) strategies.

3. Grammar – Expressing Purpose

Forms of Purpose

Structure	Example
to + verb	She studies hard to become a researcher.
in order to + verb	He joined a club in order to gain experience.
for + noun / -ing	This device is used for measuring temperature.
so that + subject + can/will	They attended the workshop so that they could improve their skills.

Practice A – Choose the Correct Option

1. She entered the physics competition (so that / in order to) challenge herself.
2. Scientists share data (to / for) increase accuracy.
3. He bought a microscope (for / so that) he could observe bacteria.

Practice B – Rewrite

1. He attends conferences. He wants to meet experts.
   → _____
2. They built a solar oven. They wanted to reduce electricity use.
   → _____

4. Speaking:

In pairs, discuss these questions:

- What scientific problem in Morocco would you like to help solve, and how?

- Do you consider science a “mission” or simply a subject?

5. Writing:

Write a paragraph on this topic:

“Science should be used for the progress of society. Do you agree? Explain with examples.”

Use at least 3 vocabulary words from the lesson and 2 purpose structures (to / in order to / so that / for + -ing).

SEMESTER 5 – UNIT 2 – LAB EQUIPMENT AND SAFETY

1. Reading Comprehension

Working Safely in the Laboratory

The laboratory is one of the most exciting and dangerous places for a science student. It is where theories are tested, reactions are observed, and discoveries are made. However, without proper safety measures, it can also become a place of serious accidents. Every student must understand that scientific progress depends not only on intelligence but also on discipline and responsibility.

Before entering the laboratory, students should always wear appropriate clothing: a lab coat, safety goggles, and closed shoes. Long hair must be tied back, and jewelry should be avoided. These precautions may seem simple, but they prevent burns, spills, and contamination. One careless moment can ruin hours of work or even harm someone.

In addition to personal safety, cleanliness and organization are essential. Equipment such as beakers, test tubes, pipettes, and microscopes must be used correctly and returned to their places after experiments. Chemical substances should never be mixed without instruction, and labels must always be read carefully. In case of an accident—such as a chemical splash or a small fire—students should immediately inform the supervisor rather than panic or try to hide the problem.

Teamwork is also part of laboratory safety. Each member of a group should have a specific role and remain alert. Communication, cooperation, and mutual respect create a safe working environment. Remember: science aims to improve life, not endanger it. A good scientist is not only curious but also cautious.

A. General Understanding

1. What is the main purpose of the text?
2. Why does the author describe the lab as both exciting and dangerous?

B. True / False / Not Mentioned

Statement	T / F / NM
Wearing safety goggles is optional.	
Students should report any accident to the supervisor.	
Jewelry helps protect students from burns.	
Teamwork contributes to safety in the lab.	

C. Multiple Choice

1. What does the author say is *as important as intelligence* in the lab?
   a) Creativity b) Responsibility

   c) Curiosity d) Speed

2. What should students do after finishing an experiment?
   a) Leave the equipment on the table
   b) Wash all materials and put them back in place
   c) Throw away used chemicals
   d) Leave immediately

D. Short Answer

1. Mention two safety rules from the text.
2. What should students do if an accident occurs?

2. Vocabulary Exercises

A. Match the Word to Its Meaning

Word	Meaning
contamination	a) glass container used for liquids
goggles	b) protection for the eyes
discipline	c) control and order in behavior
beaker	d) process of becoming impure or dirty

B. Label the Equipment

Write the correct name under each description:

1. Used to heat liquids over a flame → _____
2. Used to hold small samples → _____
3. Used to measure exact volumes → _____
4. Used to observe tiny organisms → _____

(Answer choices: test tube, microscope, pipette, Bunsen burner)

C. Word Formation

Verb	Noun	Adjective
protect	_____	_____
measure		
observe	_____	_____

3. Grammar – Countable and Uncountable Nouns + Imperatives

A. Explanation

Countable nouns → we can count (one beaker, two beakers).
Uncountable nouns → cannot be counted (water, equipment, information).

Examples:

- *We need three test tubes.* (countable)
- *There is some water in the flask.* (uncountable)
- *Don't touch the hot glass!* (imperative = direct instruction)
- *Always wear your gloves!* (imperative = safety command)

Imperatives are used to give instructions or warnings in labs.
Form: *Base verb + object* (no subject).
Examples:

- *Clean the table after the experiment.*
- *Do not mix unknown chemicals.*

B. Practice 1 – Countable or Uncountable?

Circle the correct form.

1. There is some (equipment / equipments) on the table.
2. We need more (beaker / beakers) for the experiment.
3. She spilled some (chemical / chemicals) on the floor.
4. The professor gave us useful (information / informations).

C. Practice 2 – Imperatives

Complete the safety instructions with the correct imperative form.

1. _____ (wear) your lab coat before starting.
2. _____ (not / eat) or drink in the laboratory.
3. _____ (clean) the table after finishing.
4. _____ (report) any accident immediately.

4. Speaking:

In pairs or small groups, discuss:

- What lab safety rule do you think is the most important? Why?
- Have you ever witnessed or heard of a lab accident? What happened?
- How can teamwork improve safety in the laboratory?

5. Writing:

Write a short report (150–180 words):

"A Day in the Laboratory: How We Keep It Safe and Organized."

Include:

- Description of 3–4 pieces of equipment
- 2–3 safety rules
- At least 2 **imperative sentences**
- At least 2 examples of **countable and uncountable nouns**

1. Reading Comprehension

The Power of Process Thinking in Science

Modern science is built on a clear understanding of processes—how things change, react, or develop over time. Whether we are observing the water cycle, analyzing chemical reactions, or studying photosynthesis, we rely on a logical sequence of actions and results. Process thinking allows scientists to explain not only *what* happens but also *how* and *why* it happens.

Take the process of **photosynthesis**, for example. It is not a random event but a chain of carefully ordered steps. First, light energy from the sun is absorbed by chlorophyll in the leaves. Then, this energy is converted into chemical energy that allows the plant to transform carbon dioxide and water into glucose and oxygen. The overall process is essential for life on Earth because it produces the oxygen that humans and animals breathe.

In laboratories, process-based observation is equally important. When chemists design an experiment, they must describe every stage in detail: the materials used, the procedure followed, and the results observed. Scientific accuracy depends on being able to repeat the same process and obtain the same outcome. This is why lab reports use **the passive voice**, emphasizing the process itself rather than the person performing it. For instance, instead of writing *"We heated the solution,"* scientists write *"The solution was heated to 80°C."* This focuses attention on the action and maintains objectivity.

Process thinking is not limited to natural sciences; engineers use it to design machines, and computer scientists apply it in programming algorithms. Understanding and describing processes train the mind to be systematic, patient, and precise. As the saying goes, *"Science is not only about discovery—it is about understanding how discovery happens."*

A. General Understanding

1. What is the main idea of the text?
2. Why is describing a process important in science?

B. True / False / Not Mentioned

Statement	T / F / NM
Photosynthesis converts oxygen into carbon dioxide.	
The passive voice is used to make scientific writing more objective.	
Engineers do not use process thinking.	
The author believes process thinking improves logical skills.	

C. Multiple Choice

1. The main benefit of process thinking is:
   a) Memorizing facts b) Understanding cause and effect
   c) Writing faster reports d) Avoiding experiments

2. In scientific writing, the passive voice is preferred because:
   a) It makes sentences shorter b) It focuses on the process, not the person
   c) It sounds more emotional d) It avoids repetition

D. Inference & Short Answer

1. Why is photosynthesis described as “essential for life on Earth”?
2. What qualities does process thinking help develop in scientists?
3. Find one example of the **passive voice** in the text and explain its function.

2. Vocabulary Exercises

A. Match the Word to Its Meaning

Word	Meaning
sequence	a) the result or effect of an action
convert	b) order in which events happen
outcome	c) to change something from one form to another
accuracy	d) being correct or exact
emphasize	e) to give special importance to something

B. Word Formation

Verb	Noun	Adjective
analyze	_____	analytical
react		
describe	_____	descriptive
observe	_____	_____

C. Use the Words in Context

Complete the sentences using the correct form of the words above.

1. Every experiment must include an accurate _____ of the steps.
2. The _____ of the test was different from what we expected.
3. Scientists often _____ data to identify patterns.
4. Temperature can influence how fast chemicals _____ with each other.

3. Grammar – Sequencing and Passive Voice

A. Explanation

1. Sequencing Expressions are used to describe the order of actions in a process:

First, *Then*, *After that*, *Next*, *Finally*, *At the end*, *Once*

Example: *First, the solution is mixed with water. Then, it is heated to 60°C.*

2. Passive Voice
Used when the focus is on the action, not the actor.
Form: **be + past participle**

- *The sample is analyzed in the laboratory.*
- *The mixture was heated to 100°C.*

Why use it?
In scientific writing, the process is more important than who performs it.

B. Practice 1 – Rewrite in the Passive Voice

1. The students recorded the temperature every five minutes.
   → _____
2. We cleaned the test tubes after the experiment.
   → _____
3. They will publish the results in the university journal.
   → _____
4. Scientists discovered a new element.
   → _____

C. Practice 2 – Order the Steps

Put the sentences in the correct order using sequencing words.

1. The plant produces glucose.
2. Chlorophyll absorbs sunlight.
3. Carbon dioxide and water enter the leaves.
4. Oxygen is released into the air.

→ Correct sequence: _____

D. Practice 3 – Complete Sentences

Use the words: *first, then, after that, finally*

1. _____, the materials are collected.
2. _____, the liquid is poured into the beaker.
3. _____, it is heated slowly.
4. _____, the temperature is recorded.

4. Speaking:

In groups of three:

1. Choose a simple scientific process (e.g., water cycle, digestion, distillation).
2. Prepare a short oral explanation (2–3 minutes) describing it step by step using **sequencing words** and **passive voice**.
3. One student presents; others ask clarification questions.

5. Writing:

Write a paragraph (180–200 words) describing one of the following:

- *The process of water purification*
- *The life cycle of a plant*
- *How a scientific experiment is conducted in your field*

Instructions:

- Use at least **five sequencing expressions** (first, next, etc.)
- Include at least **five passive voice structures**
- Use vocabulary from this week (e.g., sequence, convert, outcome, accuracy)

1. Reading Comprehension

How Scientists Read: The Hidden Skill Behind Discovery

Scientific reading is a skill that distinguishes a passive learner from an active researcher. In every scientific field – from molecular biology to astrophysics – the ability to understand and evaluate research articles determines how successfully one can learn, question, and innovate. Yet, many university students approach scientific texts as if they were reading a newspaper. They expect immediate clarity, simple sentences, and personal opinions. In reality, research writing demands much more: attention to structure, precision in terminology, and a critical mindset.

A scientific article typically follows a standard format known as **IMRaD**: *Introduction, Methods, Results, and Discussion.* Each section serves a unique purpose. The *Introduction* presents the background and defines the problem; the *Methods* explain how the study was conducted; the *Results* describe what was found; and the *Discussion* interprets these findings, connecting them to existing knowledge. Understanding this structure helps readers predict what type of information they will encounter, which in turn improves comprehension.

However, the challenge lies not only in structure but also in **language**. Research articles often employ complex syntax, nominalized verbs (e.g., "observation" instead of "observe"), and passive constructions that emphasize the process rather than the person. This formal tone creates objectivity but can make the text seem distant or abstract. For instance, "The samples were collected from three sites" hides the human agent, focusing solely on the scientific act. Such language choices are intentional – they reflect the discipline's values of precision, neutrality, and replicability.

Moreover, **reading critically** is as important as understanding the words. Critical readers question the author's assumptions, compare data with other studies, and evaluate whether the conclusions are supported by evidence. Blind acceptance of scientific claims can be as dangerous as ignorance itself. As one researcher once said, "Good scientists doubt everything, including their own results." This intellectual humility drives progress, since science evolves through correction, debate, and reinterpretation.

For Moroccan science students, engaging with English-language research has double value: it builds both linguistic competence and professional awareness. Around 80% of global scientific publications appear in English. Therefore, students who master scientific reading not only access the latest findings but also gain the vocabulary and analytical thinking necessary for postgraduate research or international collaboration. Scientific literacy, ultimately, is not about reading more – it is about reading *better*.

Comprehension and Critical Thinking Tasks

A. Identifying Structure

1. The text mentions the "IMRaD" structure. Match each letter with its correct description:
    - I = _____
    - M = _____
    - R = _____
    - D = _____
2. Why is recognizing this structure helpful when reading a research article?

B. Vocabulary in Context

Find words or expressions in the text that mean the following:

1. The quality of being exact and detailed: _____
2. To think carefully and form a judgment: _____
3. Words made from verbs to form abstract nouns (e.g., "observation"): _____
4. The act of checking whether something can be repeated or confirmed: _____

C. Cause and Effect

Complete the sentences using information from the text.

1. Research writing uses passive voice because _____.
2. Students who read research articles in English benefit because _____.
3. Critical reading improves science because _____.

D. Paraphrase Recognition

Choose the sentence that best expresses the same idea as the one in the text.

1. *"Good scientists doubt everything, including their own results."*
   a) Scientists should always trust their first findings.
   b) Skepticism is part of scientific progress.
   c) Scientists must never question their own methods.
   d) Doubt prevents discovery.
2. *"Scientific literacy is not about reading more – it is about reading better."*
   a) Quantity of reading is more important than quality.
   b) Students should read everything available.
   c) Deep comprehension matters more than volume.
   d) Reading less always leads to success.

E. Sequencing Ideas

Arrange these ideas in the order they appear in the text (1 – 5):

☐ The importance of English in global science
☐ The structure of scientific papers
☐ The formal language used in research writing
☐ The role of critical reading in science
☐ The difference between passive and active learners

F. Discussion Questions (Oral / Written)

1. Why do scientific texts use complex grammar and vocabulary instead of simple expressions?
2. In what ways does critical reading help prevent misinformation in science?
3. Do you agree that reading "better" is more valuable than reading "more"? Explain.

2. Vocabulary Development

A. Academic Word List

Word	Part of Speech	Definition	Example Sentence
precision	noun	exactness, accuracy	Scientific instruments must operate with great precision.
assumption	noun	a belief accepted without proof	Every hypothesis begins with an assumption.
interpret	verb	explain or understand the meaning	Students must interpret data before drawing conclusions.
neutrality	noun	the state of not showing bias	Neutrality is essential when analyzing research.
replicate	verb	to repeat an experiment to verify results	Other labs tried to replicate the experiment.

B. Word Formation – Complete the table.

Verb	Noun	Adjective
interpret	interpretation	interpretative
assume	assumption	
analyze	_____	_____
observe		
innovate	_____	_____

3. Grammar: Relative Clauses

A **relative clause** gives extra information about a noun.
It is introduced by a relative pronoun: **who, which, that, whose, where.**

Pronoun	Used For	Example
who	people	The scientist **who** led the study won an award.
which	things	The method **which** was used produced reliable results.
that	people or things	The article **that** you read is from *Nature*.
whose	possession	The student **whose** paper was selected received a grant.
where	place	The lab **where** the samples were analyzed is in Casablanca.

B. Practice 1 – Combine Sentences

1. The professor explained the experiment. He wrote the textbook we use.
   → The professor **who** _____.
2. The study was published last month. It examines renewable energy.
   → The study **that** _____.
3. The university built a new laboratory. Students can test chemical reactions there.
   → The university built a new laboratory **where** _____.
4. The researcher's data were precise. Her accuracy impressed the committee.
   → The researcher **whose** _____.

C. Practice 2 – Identify and Correct

Find and correct the error in each sentence.

1. The microscope which we buyed yesterday is broken.
2. The scientist who's theory was wrong corrected it later.
3. The journal that publish the article is based in London.

D. Practice 3 – Complete Sentences

Use relative clauses to make sentences more informative.

1. Science is a discipline _____.
2. A hypothesis is an idea _____.
3. A scientist is someone _____.
4. The data _____ were difficult to interpret.

4. Speaking

Group Task:
Each group chooses a short research article (in English) from a scientific journal or online source.

- Summarize the main idea in 2 – 3 minutes.
- Use at least **three relative clauses** and **five key vocabulary items** from today's lesson.
- Discuss: *What makes this article reliable or unreliable?*

5. Writing

A **summary** is a **short version** of a text that presents only the **main ideas** in your own words.
Follow these steps:

1. **Read** the text carefully and identify the **main argument**.
2. **Highlight** key points, not examples or statistics.
3. **Avoid copying sentences**; paraphrase the author's ideas.
4. **Keep it objective** – do not add opinions or comments.
5. **Use connectors** (e.g., firstly, moreover, finally) and concise language.
6. **Check** that your summary is coherent and clear.

Write a **200-word analytical summary** of the text *"How Scientists Read: The Hidden Skill Behind Discovery."*

Your summary should:

- Identify the text's **main argument and structure (IMRaD)**
- Include **three relative clauses**
- Use **vocabulary** such as *assumption, interpret, precision, replicate,* and *neutrality*
- Conclude with your opinion on why critical reading is essential for scientific success

SEMESTER 5 – UNIT 5 – Graphs and Data Interpretation

Why Scientific Decisions Depend on Data Literacy

Over the last twenty years, the scientific community has entered what many researchers call *the age of data dependence*. With the expansion of digital sensors, satellite imaging, laboratory automation, and online databases, scientists are no longer limited by a lack of information. Instead, they are challenged by the difficulty of analysing enormous and complex datasets. In fields such as environmental science, epidemiology, and urban planning, data literacy—not simply reading numbers, but interpreting their meaning – has become a core scientific competence.

For science students, developing strong data-interpretation skills is essential. A graph is not just a visual decoration; it is a simplified model of reality. Students must learn to examine several features before accepting the message a graph appears to show. First, they should **read the axes carefully**, paying attention to units, time intervals, and measurement scales. Even small changes in scale can dramatically alter the appearance of a trend. Second, students must **identify the variables** represented and determine whether they are continuous (like temperature) or categorical (like types of pollutants). This affects which type of graph is appropriate.

Science students should also be able to **distinguish correlation from causation**. Two lines that rise together do not necessarily indicate that one variable causes the other to change. Careful readers must look for missing variables, unrecorded factors, or gaps in the data. In addition, students should learn to question **how the data was collected**. Was it gathered evenly across regions? Was the sample size sufficient? Were instruments calibrated properly? Understanding these aspects helps prevent misinterpretation.

Graphs and charts simplify complex information, but they also shape the way readers perceive it. A line graph showing annual pollution levels in Morocco's industrial zones might suggest a steady increase, but a closer look at the vertical scale may reveal that the change is relatively small. Likewise, a bar chart comparing population densities across Moroccan regions highlights differences, but these differences depend heavily on measurement units and data-collection methods. As a result, scientists must not only interpret visual patterns but also question the reliability and limitations of the data.

Another challenge is the growing use of predictive models. These models rely on large datasets to estimate future trends—such as coastal erosion, rainfall variability, or population expansion. While such models are powerful, they can be misleading if the underlying data is incomplete or biased. For instance, rural areas may lack consistent data collection, causing models to emphasise urban trends. This can lead policymakers to make decisions that favour well-documented regions while neglecting less-studied ones.

Authentic data literacy therefore requires a combination of skills: understanding how data is collected, knowing how to visualise it effectively, interpreting patterns critically, and recognising uncertainty. Science students must approach graphs as analytical tools rather than simple illustrations. Readers should ask: *What does this graph show? What does it hide? What assumptions does it make? What alternative explanations exist?* Such reflective questioning is central to scientific thinking.

In Morocco, as in many countries, strengthening data literacy is vital for informed decision-making. Whether the goal is to confront environmental pressures, manage water resources, or plan sustainable urban growth, policymakers and scientists depend on accurate interpretations of visual data. For science students, this means approaching every graph with a critical and questioning mindset – recognising what the data reveals, what it might conceal, and how confidently conclusions can be drawn. Developing these habits early prepares students to contribute responsibly to evidence-based scientific work and national development.

I. Comprehension

1. Multiple Choice – Global Understanding

Choose the best answer:

The main purpose of the text is to show that data literacy is:

a) Only necessary for mathematicians.
b) Mainly about learning computer programming.
c) A fundamental skill for interpreting scientific information.
d) Less important than collecting large amounts of data.

2. Short Answer – Key Details

Answer in one or two sentences.

1. What two features of a graph must students check before accepting its message?
2. How can gaps in data influence scientific conclusions?

3. Table Completion – Types of Problems Mentioned

Fill in the table with information from the text.

Problem Type	Example from Text
Graph Design Issue	_____
Data Collection Issue	
Interpretation Issue	_____

4. Sentence Completion – Understanding Cause/Effect

Complete the sentences using information from the text.

1. Predictive models may become misleading when _____.
2. A trend on a graph might look stronger than it really is because _____.

5. True/False/Not Mentioned

1. All Moroccan regions collect scientific data with equal regularity.
2. A line graph may exaggerate or minimise a trend depending on its scale.
3. The text states that correlation always indicates causation.
4. Students should examine how data was collected before trusting a graph.

6. Inference – Reading Between the Lines

1. What does the author suggest about the reliability of visual patterns alone?
2. Why might science students need to question what a graph *hides*?

7. Critical Reflection

Imagine you are a policymaker using limited data. What risks could arise if you base decisions only on urban datasets?

8. Application Task – Real-World Problem Solving

You receive a bar chart showing a sudden rise in air-pollution levels. List **three** checks you should make before accepting the result.

II. Vocabulary Development

1. Match the terms with their definitions:

1) categorical variable _____
2) correlation _____
3) dataset _____
4) measurement scale _____
5) calibration _____
6) predictive model _____

a) A collection of related data points, usually organised in tables, files, or databases so they can be analysed. *Example:* A spreadsheet containing the monthly rainfall for each Moroccan city is a dataset.
b) The process of adjusting or fine-tuning an instrument or measurement tool so that it gives correct and reliable results. *Example:* A scale must be calibrated before weighing chemicals in a laboratory.
c) A type of variable that represents groups, categories, or labels rather than numerical values. *Example:* "Type of smartphone" (Samsung, Apple, Huawei) is a categorical variable.
d) A statistical relationship between two variables that shows how they change together – either increasing, decreasing, or showing no clear pattern. *Example:* There is a strong correlation between hours studied and exam scores.
e) A mathematical or statistical system used to estimate or forecast future outcomes based on existing data. *Example:* Weather apps use predictive models to forecast tomorrow's temperature.
f) A defined range, unit, and spacing used to measure and interpret numerical data on instruments or graphs. *Example:* A thermometer with markings every 1°C uses a specific measurement scale.

2. Word-Formation Practice:

Use the following words to form a noun, adjective, or verb that completes each sentence.

interpret – dependent – reliable – visual – analyse – collection

1. Strong data _____ is essential for scientific work.
2. The graph provides a clear _____ summary of the results.
3. Scientists must question the _____ of data sources.
4. Moroccan researchers are increasingly _____ on satellite data.
5. Students learn to _____ graphs more accurately with practice.
6. Poor data _____ methods can affect scientific conclusions.

III. Grammar – Using Comparative Structures in Scientific Writing

A. Rewrite the sentences using the comparative form:

1. Urban pollution levels are high, but industrial-zone pollution levels are even higher.
   → Industrial-zone pollution levels are _____ urban pollution levels.
2. The 2024 model is efficient, but the 2025 model is efficient too.
   → The 2025 model is _____ the 2024 model.
3. Rural data coverage is limited, and coastal data coverage is extensive.
   → Coastal data coverage is _____ rural data coverage.

B. Complete each statement with an appropriate comparative expression (more / less / fewer + noun):

1. Regions with _____ monitoring stations often produce incomplete datasets.
2. A graph becomes clearer when it has _____ unnecessary labels.
3. Students make _____ mistakes when they read axes carefully.

IV. Speaking – Group Debate: "Can Data Ever Be Neutral?"

Form groups of 3 – 4 students. Read the discussion prompt below, then debate the questions.
Each student should take a position, defend it with examples, and respond to others' arguments.

Scientists often treat data as objective, but every dataset is influenced by choices – what to measure, who to include, which variables to ignore, and how the results are visualised. This raises an important question: *Is any data truly neutral, or is it always shaped by human decisions?*

Debate Questions:

1. To what extent can data be considered objective?
2. Who decides what counts as "important data"?
3. Can a graph or chart be accurate but still misleading?
4. Is it ever acceptable to simplify data for the public?
5. What responsibilities do scientists have when presenting results to society?
6. Can two scientists interpret the same dataset differently and both be right?
7. How can students develop "healthy skepticism" when reading scientific graphs?

Groups summarise their main points and present a **short collective statement** addressing the central question: **"Can Data Ever Be Neutral?"**

V. Writing – Analytical Paragraph: "Describe and Interpret a Scientific Graph"

Choose a graph and write a **7 – 10 sentence analytical paragraph**. Your paragraph must:

- Identify the type of graph and the variables shown.
- Describe the main trend(s) accurately.
- Include **at least two comparative structures**.
- Comment on data reliability, scale, or missing information.
- End with a cautious statement about the confidence level of your interpretation.

Use clear scientific language and avoid simply listing numbers. Focus on interpretation, not transcription.

SEMESTER 5 – UNIT 6 – Environmental Issues in Morocco

Plastic Pollution and Environmental Challenges in Agadir

Agadir, a rapidly growing coastal city in Morocco, is facing significant environmental challenges. Among these, plastic pollution has emerged as a major concern. Single-use plastics – bags, bottles, packaging – litter streets, beaches, and waterways, creating hazards for wildlife, fisheries, and public health. Local rivers often carry plastic debris to the Atlantic Ocean, threatening marine ecosystems and affecting communities dependent on fishing.

Plastic pollution in Agadir is not merely a local issue; it is part of a wider environmental crisis in Morocco. Studies indicate that plastic production in the country has risen steadily over the last decade, mirroring global trends. Improper waste management, combined with population growth and tourism, exacerbates the problem. Inadequate recycling infrastructure means that a large proportion of plastics ends up in landfills or the natural environment. This leads to soil contamination, disruption of aquatic life, and long-term ecological consequences.

The Moroccan government has implemented policies to reduce single-use plastics, including bans on plastic bags and campaigns to raise public awareness. While these measures have had some success, the effectiveness remains debated. Enforcement is uneven, and alternatives – such as reusable bags – are not always affordable or widely available. Moreover, public compliance varies depending on education and socioeconomic factors. The issue is further complicated by the fact that plastic pollution is intertwined with consumer habits, industrial packaging practices, and tourism, all of which influence the environment directly and indirectly.

Understanding environmental issues in Agadir requires not only identifying the sources of pollution but also analysing their consequences. Cause-and-effect thinking is essential: plastic litter leads to blocked drainage systems, which in turn cause urban flooding during the rainy season; plastic in rivers and oceans harms fish populations, which affects the livelihoods of local fishermen; and microplastics in soil and water can accumulate in the food chain, posing health risks. These cascading effects highlight the importance of systematic environmental management and public engagement.

For science students, learning to map causes and effects, evaluate interventions, and consider both local and broader contexts is crucial. Morocco's efforts to reduce plastic pollution demonstrate the complexity of environmental governance: solutions must balance ecological sustainability, economic feasibility, and social behaviour. By examining Agadir as a case study, students can develop critical thinking skills, understand the interconnections between human activity and environmental outcomes, and propose evidence-based interventions.

A. Comprehension

1. Multiple Choice – Main Idea

1. What is the main focus of the text?
   a) The history of plastic production in Morocco
   b) Environmental challenges caused by plastic pollution in Agadir
   c) Tourism development in Morocco
   d) Marine biodiversity in the Atlantic Ocean
2. Why does the text describe cause-and-effect relationships?
   a) To show that all pollution is reversible
   b) To demonstrate the interconnected impact of plastic on the environment and society

c) To list government policies only
d) To explain plastic production methods

2. Short Answer – Key Details

1. What are the main sources of plastic pollution in Agadir?
2. Name two ecological or social effects of plastic pollution mentioned in the text.
3. Mention one reason why plastic bans are not fully effective in Morocco.

3. True / False / Not Mentioned

1. Plastic pollution only affects marine animals.
2. Plastic production in Morocco has decreased over the last decade.
3. Enforcement of plastic bans is consistent throughout the country.
4. Microplastics can enter the food chain.

4. Cause-and-Effect Mapping

Fill in the chain of effects from the text:

Plastic litter → _____ → _____ → _____

B. Vocabulary Development

1. Match the Terms with Definitions

a. **single-use plastics**
b. **microplastics**
c. **infrastructure**
d. **compliance**
e. **ecological consequence**
f. **cascading effect**

Definitions:

1. Very small pieces of plastic that result from the breakdown of larger items, often entering water and soil.
2. Plastic items designed to be used only once before disposal, such as bags and bottles.
3. The physical and organisational structures and facilities needed for waste collection, recycling, and disposal.
4. The act of following rules, regulations, or guidelines.
5. Environmental outcomes that affect ecosystems, such as species loss or soil contamination.
6. A series of events in which one effect triggers another, leading to multiple consequences.

2. Word Formation Exercise

Use the following words to form nouns, adjectives, or verbs:

pollute – recycle – enforce – sustainable – contaminate – manage

1. Plastic waste continues to _____ rivers and beaches.
2. Governments must improve waste _____ to reduce pollution.

3. Public _____ with environmental laws is crucial for success.
4. Scientists research the _____ impact of microplastics on marine life.
5. Communities are encouraged to _____ packaging materials.
6. Policies aim to promote _____ development in urban planning.

C. Grammar Focus – Cause and Effect Structures

1. Linking Words / Phrases

Fill in with **because, due to, as a result, therefore, which leads to, consequently**:

1. Urban flooding occurs _____ blocked drainage systems.
2. Microplastics enter the food chain _____ pollution in rivers.
3. Enforcement is uneven, _____ plastic pollution remains high.
4. Recycling infrastructure is inadequate, _____ a large proportion of plastic ends up in the environment.

2. Sentence Transformation

Rewrite using a cause-effect connector:

1. Plastic pollution harms fish populations. This affects the livelihoods of local fishermen.
   → _____
2. Tourism increases plastic waste. Beaches become littered.
   → _____

D. Speaking / Debate Activity

Debate Topic: "Plastic Ban in Morocco – Effective or Not?"

Instructions:

- Form two groups: **Pro-ban** vs **Critical perspective**.
- Each group prepares arguments using evidence from the text.

Outcome: Each group summarises its position in 3 – 4 sentences. Be ready to respond to the other side's arguments.

E. Writing – Opinion Paragraph

Title: "Plastic Bans in Morocco: Effective Environmental Solution or Limited Measure?"

Task: Write an **opinion paragraph** presenting your viewpoint. Include:

- A clear opinion (agree, partially agree, disagree).
- **Two cause-effect links** from the text.
- **At least two comparative structures** (e.g., *more effective than*, *less harmful than*).
- Reference at least one example from Agadir or Morocco.
- Conclude with a recommendation or reflection.